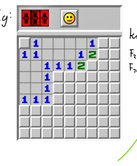
Minesweeper

Samstag, 6. Dezember 2025 12:33

Input: $\underbrace{n, m}_{\text{dimension of your field}}$, $\underbrace{b}_{\text{count of bombs}}$, Given field → Eg:



$n = 9 \quad m = 9 \quad b = 70$

Known variable that are true:

$F_{11}\,7, F_{12}\,0, F_{13}\,0, F_{14}\,0, F_{15}\,0, F_{16}\,0, F_{17}\,7$

$F_{30}\,7, F_{31}\,7, F_{32}\,0 \ldots\ldots$

Output: Is given board solvable?

= Is there an valid assignment of bombs?

Definition / Explanation of variable

$F_i X \rightarrow$ Specific field on your board with $1 \le i \le n \cdot m$

With a certain value $X \in \{\underbrace{0,1,2,3,4,5,6,7,8}_{\text{number}}, \underbrace{9}_{\text{bomb}}\}$

F_1	F_2	$\cdots$	F_m
F_{m+1}	F_{m+2}	$\cdots$	F_{2m}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$F_{(n-1)\cdot m+1}$	$\cdots$	$\cdots$	$F_{n\cdot m}$

$\Big\}\, m$

$\underbrace{\qquad\qquad}_{n}$

Neighbours: We define the 8+9 neighbours of F_i as $n\,7$ to $n\,8$

n_K If $n_K = True \rightarrow$ neighbour is a bomb

If $n_K = False \rightarrow$ neighbour is not a bomb

This is just for simplicity for iteration over the neighbours and to not always have to write „i" to determinate the bomb value

F_{i-m-1}	F_{i-m}	F_{i-m+1}
F_{i-1}	F_i	F_{i+1}
F_{i+m-1}	F_{i+m}	F_{i+m+1}

→

$n\,7$	$n\,2$	$n\,3$
$n\,4$	F_i, j	$n\,5$
$n\,6$	$n\,7$	$n\,8$

Note: all neighbours that are outside the your board always have the value 0

→ $F_i\ 0 = True \quad \forall i < 1$ or $i > n \cdot m$

$F_i\ 1 \ldots F_i\ 7 = False$

Transition from all your rules to SAT

Rule: Each field has at least one value

$$\bigwedge_{i=1}^{n\cdot m} \left(\bigvee_{x=0}^{9} F_{ix} X \right)$$

$(F_{11} 0 \vee F_{11} 1 \vee F_{11} 2 \vee \ldots) \wedge$

$(F_{12} 0 \vee F_{12} 1 \vee \ldots\ldots \quad) \wedge$

$\vdots$

$(F_{n\cdot m} 0 \vee F_{n\cdot m} 1 \vee \ldots\ldots \quad)$

$n \cdot m$ clauses

10 literal each

Rule: Each field has at most one value

$$\bigwedge_{i=1}^{n\cdot m} \bigwedge_{0 \le x_1 < x_2 \le 9} \left(\overline{F_{i_1} X_1} \vee \overline{F_{i_2} X_2} \right)$$

$(\overline{F_{11} 0} \vee \overline{F_{11} 1}) \wedge (\overline{F_{11} 0} \vee \overline{F_{11} 2}) \wedge \ldots$

$\vdots$

$(\overline{F_{nm} 0} \vee \overline{F_{nm} 1}) \ldots (\overline{F_{nm} 8} \vee \overline{F_{nm} 9})$

$n \cdot m \cdot 36$ clauses

2 literal each

Rule: Across the board there are at least b bomb

$j = n \cdot m - b + 1$

$$\bigwedge_{1 \le i_1 < i_2 < \ldots < i_j \le n \cdot m} \left(F_{i_1} 9 \vee F_{i_2} 9 \vee \ldots \vee F_{i_j} 9 \right)$$

$\binom{n \cdot m}{j}$ clauses

j literal each

Rule: Across the board there are at most b bomb

$$\bigwedge_{1 \le i_1 < i_2 < \ldots < i_b < i_{b+1} \le n \cdot m} \left(\overline{F_{i_1} 9} \vee \overline{F_{i_2} 9} \vee \ldots \vee \overline{F_{i_b} 9} \vee \overline{F_{i_{b+1}} 9} \right)$$

$\binom{n \cdot m}{b+1}$ clauses

b+1 literal each

Rule: if a field has value 0, no neighbour is a bomb → n_K is false

$$\bigwedge_{i=1}^{n\cdot m} \bigwedge_{k=1}^{8} \left(\overline{F_i 0} \vee \overline{n_K} \right)$$

$n \cdot m \cdot 8$ clauses

2 literal each

Rule: If a field has value 1, exactly one neighbour is a bomb → $n_K 1 = True$

at least one: $\bigwedge_{i=1}^{n\cdot m} \left(\overline{F_i 1} \vee n_1 \vee n_2 \vee n_3 \vee \ldots \vee n_8 \right)$ — $n \cdot m$ clauses, 9 literal each

at most one: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < k_2 \le 8} \left(\overline{F_i 1} \vee \overline{n_{k_1}} \vee \overline{n_{k_2}} \right)$ — $n \cdot m \cdot 28$ clauses, 3 literal each

Rule: If a field has value 2, exactly 2 neighbours are bombs

at least two: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 2} \vee n_{k_1} \vee \ldots \vee n_{k_j} \right)$ — $n \cdot m \cdot 8$ clauses, 9 literal each

$j = 7 \quad (8 - X + 1)$

at most two: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < k_2 < k_3 \le 8} \left(\overline{F_i 2} \vee \overline{n_{k_1}} \vee \overline{n_{k_2}} \vee \overline{n_{k_3}} \right)$ — $n \cdot m \cdot 56$ clauses, 4 literal each

Rule: If a field has value 3, exactly 3 neighbours are bombs

at least 3: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 3} \vee n_{k_1} \vee \ldots \vee n_{k_j} \right)$ — $n \cdot m \cdot 28$ clauses, 7 literal each

$j = 6$

at most 3: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 3} \vee \overline{n_{k_1}} \vee \ldots \vee \overline{n_{k_j}} \right)$ — $n \cdot m \cdot 70$ clauses, 5 literal each

$j = 4 \quad (X+1)$

Rule: If a field has value 4, exactly 4 neighbours are bombs

at least 4: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 4} \vee n_{k_1} \vee \ldots \vee n_{k_j} \right)$ — $n \cdot m \cdot 56$ clauses, 6 literal each

$j = 5$

at most 4: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 4} \vee \overline{n_{k_1}} \vee \ldots \vee \overline{n_{k_j}} \right)$ — $n \cdot m \cdot 56$ clauses, 6 literal each

Rule: If a field has value 5, exactly 5 neighbours are bombs

at least 5: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 5} \vee n_{k_1} \vee \ldots \vee n_{k_j} \right)$ — $n \cdot m \cdot 70$ clauses, 5 literal each

$j = 4$

at most 5: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 5} \vee \overline{n_{k_1}} \vee \ldots \vee \overline{n_{k_j}} \right)$ — $n \cdot m \cdot 28$ clauses, 7 literal each

$j = 6$

Rule: If a field has value 6, exactly 6 neighbours are bombs

at least 6: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 6} \vee n_{k_1} \vee n_{k_2} \vee n_{k_3} \right)$ — $n \cdot m \cdot 56$ clauses, 4 literal each

$j = 3$

at most 6: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 6} \vee \overline{n_{k_1}} \vee \ldots \vee \overline{n_{k_j}} \right)$ — $n \cdot m \cdot 8$ clauses, 8 literal each

$j = 7$

Rule: If a field has value 7, exactly 7 neighbours are bombs

at least 7: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 7} \vee n_{k_1} \vee n_{k_2} \right)$ — $n \cdot m \cdot 28$ clauses, 7 literal each

$j = 2$

at most 7: $\bigwedge_{i=1}^{n\cdot m} \left(\overline{F_i 7} \vee n_1 \vee n_2 \vee \ldots \vee n_8 \right)$ — $n \cdot m$ clauses, 9 literal each

Rule: If a field has value 8, all neighbours are bombs

$$\bigwedge_{i=1}^{n\cdot m} \bigwedge_{k=1}^{8} \left(\overline{F_i 8} \vee n_K \right)$$

$n \cdot m \cdot 8$ clauses

2 literal each

for $1 \le x \le 7$

at least x bomb: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i X} \vee n_{k_1} \vee \ldots \vee n_{k_j} \right)$ — $n \cdot m \cdot \binom{8}{j}$ clauses, $j+1$ literal each

$j = 8 - x + 1$

at most x bomb: $\bigwedge_{i=1}^{n\cdot m} \bigwedge_{1 \le k_1 < \ldots < k_j \le 8} \left(\overline{F_i 6} \vee \overline{n_{k_1}} \vee \ldots \vee \overline{n_{k_j}} \right)$ — $n \cdot m \cdot \binom{8}{j}$ clauses, $j+1$ literal each

$j = x + 1$

Count of all clauses

$$n \cdot m + n \cdot m \cdot 36 + \underbrace{\binom{n \cdot m}{n \cdot m - b + 1}}_{= \binom{n \cdot m}{b+1}} + \binom{n \cdot m}{b+1} + 2 \cdot n \cdot m \cdot \underbrace{\binom{8}{j}}_{x = 7 \le j \le 1} + \underbrace{n \cdot m \cdot 8}_{x = 0} + \underbrace{n \cdot m \cdot 8}_{x = 8}$$

$$= n \cdot m \cdot \left(1 + 36 + 2 \cdot \binom{n \cdot m}{b+1} + 8 + 8 \right)$$

$$= n \cdot m \cdot \left(55 + 2 \cdot \frac{(n \cdot m)!}{(b+1)! \cdot ((n \cdot m)! - (b+1))!} \right)$$

SAT clauses

$O(N) \rightarrow$ at most polynomial ✓

certificate is in P